WATCH SERVICE CENTER – SERVICE BOOKING & REPAIR TRACKING SYSTEM

A PROJECT REPORT

Submitted by

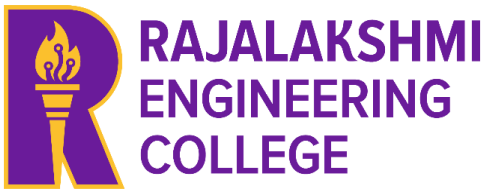
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in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY
COMPUTER SCIENCE AND BUSINESS SYSTEMS



RAJALAKSHMI ENGINEERING COLLEGE
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OCTOBER 2026

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BONAFIDE CERTIFICATE

Certified that this project report titled “Watch Service Center – Service Booking & Repair Tracking System” is the bonafide work of THARUN KRISHNA SHANMUGA SUNDAR (241401118), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester IV, under my supervision.

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EXTERNAL EXAMINER

DECLARATION

I, THARUN KRISHNA SHANMUGA SUNDAR (241401118), hereby declare that the project report titled “Watch Service Center – Service Booking & Repair Tracking System”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its form, reports, page, Deluge script and workflow — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet or external resource used in this work has been duly acknowledged in the report. AI-based tools were used to assist in preparing the report text and design diagrams, and this assistance is acknowledged here.

Place: Chennai
Date: 02 November 2026

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THARUN KRISHNA SHANMUGA SUNDAR

ABSTRACT

Watch service centres commonly record customer details, watch information and repair progress in paper registers and diaries. This makes it hard to confirm a booking to the customer, to know in which stage each watch is, and to see how many units are in the workshop or waiting to be collected.

To address these problems, a Watch Service Center application was designed, built and published on Zoho Creator. A Service Booking form captures customer name, e-mail, address, phone number, watch brand and model, issue description, serial number, dropping or collecting date and time, and a five-stage repair status. A form workflow with a Deluge sendmail script sends an automatic confirmation e-mail on every booking. A list report, a Kanban Repair Pipeline and a Workshop Calendar present the bookings in different ways, and an Analytics Dashboard page shows total units, units ready to collect and the status distribution.

The application was tested with seven sample service records covering all five repair stages. The booking form, confirmation e-mail, reports and dashboard behaved as expected, and the remaining improvements — such as sending the e-mail to the customer's own address and adding a customer portal — are documented as future enhancements.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Kanban; Service Management

PROJECT AT A GLANCE

Attribute	Details
Project Title	Watch Service Center – Service Booking & Repair Tracking System
Application Domain	Customer Service Management – Watch Repair and Servicing
Student Name & Register No.	THARUN KRISHNA SHANMUGA SUNDAR – 241401118
Zoho Creator Edition	Trial plan
Application Name (Link Name)	watch-service-center
Published / Live URL	https://creatorapp.zoho.in/tharunkrishna.2024.csbs_r/watch-service-center#Form:Service_Booking
Demo Credentials (per role)	Application shared with the examiner through the Zoho Creator Share option (owner account: tharunkrishna.2024.csbs_r)
Number of Forms / Subforms	1 / 0
Number of Reports (by type)	List: 1 Kanban: 1 Calendar: 1 Pivot/Chart: 1 (pie chart on dashboard) Others: 0
Number of Pages / Dashboards	1
Number of Workflows	Form: 1 Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	1 (Deluge sendmail script in the form workflow)
Integrations	Zoho Mail (sendmail task)
Roles & Profiles Defined	Administrator (application owner); Staff and Customer roles designed in the permission matrix
Environments Used	Zoho Creator builder (development) and live mode
Date of Deployment	01 November 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR / NFR	Functional / Non-Functional Requirement
HTML	HyperText Markup Language
KPI	Key Performance Indicator
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
REST	Representational State Transfer
UAT	User Acceptance Testing
UI	User Interface
UML	Unified Modeling Language

CHAPTER 1
INTRODUCTION

1.1 Background

Watch repair and servicing centres handle a steady flow of valuable and often sentimental items — from everyday quartz watches to premium mechanical and luxury timepieces. Each watch passes through several stages inside the workshop: it is checked in at the counter, diagnosed by a technician, sometimes held back while spare parts are ordered, worked upon at the bench and finally returned to the customer. In many small and medium service centres this journey is still recorded in paper registers, notebooks or scattered spreadsheets.

Such manual record keeping makes it difficult for the front desk to answer a simple customer question — "where is my watch right now?" — and gives the workshop manager no consolidated view of the workload. Low-code platforms such as Zoho Creator make it practical to replace these manual practices with a purpose-built application without the cost and time of conventional software development.

1.2 Problem Statement

The service centre studied in this project relies on manual booking and follow-up, which leads to the following pain points:

- Customer and watch details are written in registers, so records are easily lost, duplicated or illegible.
- There is no live view of which stage each watch is in (checked in, under diagnosis, awaiting parts, on the bench or ready), so status enquiries are answered from memory or by walking to the workshop.
- Customers receive no written confirmation after handing over a watch for service.
- Drop-off and collection appointments are noted in a diary, which makes it hard to see how many watches are due on a given day.
- Management has no quick overview of the total number of units in the workshop or the number waiting to be collected.

1.3 Objectives

- To design a service booking form that captures customer, contact, address, watch and appointment details with mandatory-field validation.
- To track every watch through the five repair stages — Checked In, Under Diagnosis, Awaiting Parts, On the Bench and Ready for Pickup — using a Kanban repair pipeline.
- To automate a confirmation e-mail on every successful booking using a Deluge script inside a form workflow.

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- To provide a workshop calendar of drop-off / collection dates and an analytics dashboard with total units, ready-to-collect count and status distribution.
- To build and publish the complete application on Zoho Creator using low-code techniques.

1.4 Scope of the Project

The scope covers service booking, repair status tracking, workshop calendar, an analytics dashboard and automated e-mail confirmation for a watch service centre. The following are outside the scope of the current release: payment collection and invoicing, spare-parts inventory, estimated-cost quotation, SMS / WhatsApp notifications, and a customer self-service login portal. Several of these are listed as future enhancements in Chapter 9.

1.5 Justification for a Low-Code Approach

Zoho Creator allows the data model, forms, reports, dashboards and workflows to be configured visually, while Deluge scripting is used only where custom logic — such as composing and sending an e-mail — is required. The platform provides cloud hosting, built-in report types (list, Kanban, calendar), responsive device previews and secure access without the need to set up servers or write a front-end. For a single-branch service centre this reduces development effort from weeks to days and keeps the application easy to modify as the business changes.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

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CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing process the customer walks in (or telephones) with a watch, and the counter staff note the customer name, contact number, the watch brand and the problem in a register. The watch is tagged and sent to the workshop. When the customer enquires about progress, the staff verbally check with the technician. Collection dates are written in a diary and customers are informed by phone call when the watch is ready.

2.2 Limitations of the Existing System

- Information is scattered across registers and diaries, with no single searchable record.
- No visibility of repair status; every enquiry interrupts the workshop.
- No written acknowledgement to the customer at the time of check-in.
- No consolidated count of units in the workshop, units awaiting parts or units ready for collection.
- High dependency on individual staff memory; risk of missed or delayed customer communication.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development effort	Weeks of design, coding, integration and testing	Days — forms, reports and workflow configured visually
Skills required	Full-stack development (front-end, back-end, database, e-mail service)	Zoho Creator builder and basic Deluge scripting
Hosting & infrastructure	Self-managed server, database and mail setup	Managed Zoho cloud platform
Reports & dashboards	Custom UI and charting libraries to be coded	Built-in list, Kanban, calendar reports and dashboard page builder
Mobile access	Separate responsive design or native app	Responsive UI with device preview and mobile app support
Security & access control	Custom authentication and authorisation	Built-in Zoho login, roles and permissions
Change management	Code changes, rebuild and redeploy	Edit the form or report in the builder and publish
Customisation limits	Unrestricted	Platform and Deluge execution limits
Cost	High initial infrastructure and developer cost	Subscription-based; low initial cost

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2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Gartner (2023)	Magic Quadrant for Enterprise Low-Code Application Platforms	Low-code platforms are positioned as a way to deliver business applications faster using visual development.	Supports the choice of a low-code platform for a small service-management application.
2	Sanchis, García-Perales, Fraile & Poler (2020)	Low-Code as Enabler of Digital Transformation in Manufacturing Industry, Applied Sciences 10(1)	Low-code allows business users and small teams to digitise processes with less dependence on traditional programming.	Validates using low-code to digitise a manual workshop process.
3	Anderson, D. J. (2010)	Kanban: Successful Evolutionary Change for Your Technology Business	Visualising work items in stage columns makes the flow and bottlenecks of work visible.	Basis for the Repair Pipeline Kanban report grouped by repair status.
4	Zoho Corporation (2026)	Zoho Creator Help Documentation	Describes forms, report types, page builder and workflow configuration.	Primary reference for building the forms, reports and dashboard.
5	Zoho Corporation (2026)	Deluge Script Reference Guide	Documents the Deluge language and the sendmail task.	Used to write the confirmation e-mail script.

2.5 Emerging Trends in Low-Code Development

Current trends in low-code development include embedding AI assistants in the application builder, event-driven automation, and ready-made connectors to messaging and payment services. For service businesses this points toward proactive customer communication — for example, automatic status messages when a repair moves to the next stage — which is identified as a future enhancement of this project.

2.6 Proposed System

The proposed Zoho Creator application digitises the complete service booking journey. A single Service Booking form captures customer and watch details; a form workflow sends an automated confirmation e-mail; a Kanban report shows every watch in its repair stage; a calendar shows appointments by date; and an analytics dashboard summarises total units and units ready to collect.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Customer	Watch owner who hands over a watch for servicing or replacement	Provides personal and watch details; receives the booking confirmation e-mail; collects the watch when it is ready.
Staff / Admin	Service desk staff, workshop technicians and the workshop manager	Creates and edits service bookings, moves watches through repair stages, checks the calendar and monitors the dashboard.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall capture customer name, e-mail, address, phone number, watch brand and model, issue description and serial number in a Service Booking form.	Staff / Admin	High
FR-02	The system shall enforce mandatory entry of customer name, e-mail and phone number before a booking can be submitted.	Staff / Admin	High
FR-03	The system shall record the dropping or collecting date and time of each booking.	Staff / Admin	High
FR-04	The system shall track each watch through five statuses: Checked In, Under Diagnosis, Awaiting Parts, On the Bench and Ready for Pickup.	Staff / Admin	High
FR-05	The system shall send an automated confirmation e-mail on every successful booking, quoting the watch brand and model.	System	High
FR-06	The system shall list all bookings in a tabular report.	Staff / Admin	Medium
FR-07	The system shall display bookings as cards grouped by status in a Kanban repair pipeline.	Staff / Admin	Medium
FR-08	The system shall display bookings by date in a workshop calendar with month, week and day views.	Staff / Admin	Medium
FR-09	The system shall present an analytics dashboard with total units, units ready to collect and a status distribution chart.	Staff / Admin	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	Forms and reports shall be simple enough for counter staff to use without training; status values shall be chosen from a fixed list.

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NFR ID	Category	Requirement
NFR-02	Security	Application access shall require Zoho authentication; personal fields (name, e-mail, address, phone) shall be flagged as personal data.
NFR-03	Responsiveness	The application shall be usable on desktop, tablet and mobile layouts supported by Zoho Creator.
NFR-04	Availability	The application shall be hosted on the Zoho cloud, with availability governed by the Zoho service.
NFR-05	Maintainability	Business logic shall be kept in small, readable Deluge scripts and configuration so that fields and statuses can be changed easily.
NFR-06	Performance	Reports, Kanban and calendar views shall load promptly for the data volume of a single service centre.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (trial plan, Indian data centre — creatorapp.zoho.in)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Microsoft Edge / Google Chrome / Mozilla Firefox (latest version)
Other Zoho services	Zoho Mail (through the Deluge sendmail task)
Mobile	Zoho Creator mobile app or any mobile browser; confirmation e-mail read in the Gmail mobile app
Client hardware	Intel Core i3 / i5 or above, 8 GB RAM, internet connection

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3.5 Use Case Model

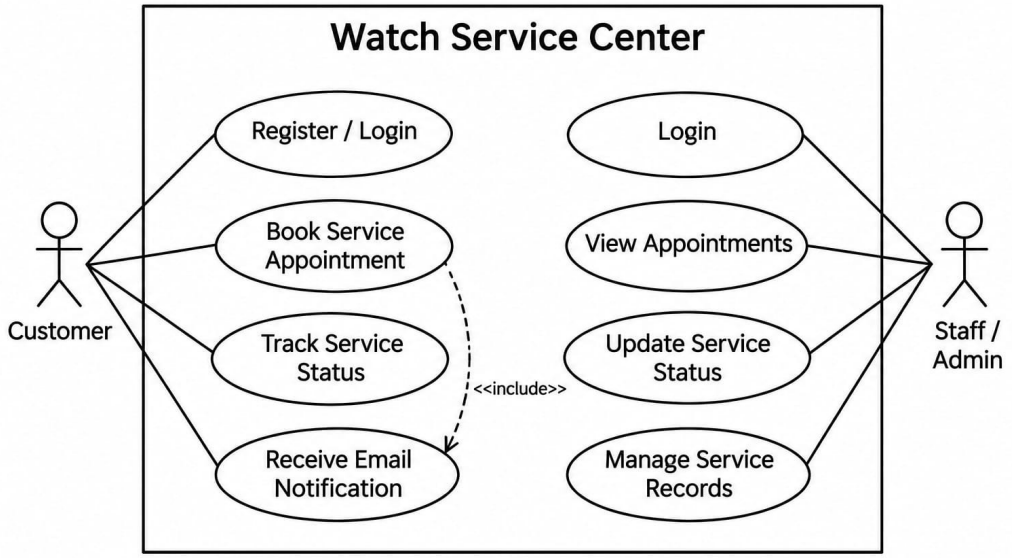


Figure 3.1: Use Case Diagram of the Watch Service Center

The use case model identifies two actors. The Customer registers or logs in, books a service appointment, tracks the service status and receives an e-mail notification (included in the booking use case). The Staff / Admin logs in, views appointments, updates the service status and manages service records. The two principal use cases are described below.

Table 3.5: Use Case Description – Book Service Appointment

Field	Description
Use Case ID	UC-01
Use Case Name	Book Service Appointment
Actor(s)	Customer (through counter staff) / Staff
Pre-condition	The user has access to the Watch Service Center application.
Main Flow	1. User opens the Service Booking form. 2. User enters customer name, e-mail, address, phone, watch brand and model, issue description, serial number, date and time. 3. User submits the form. 4. The record is saved and the message “Data Added Successfully!” is shown. 5. The “Send Welcome Email” workflow sends the confirmation e-mail.
Alternate Flow	If a mandatory field (name, e-mail or phone) is blank, the form is not submitted and the field is highlighted.
Post-condition	The booking appears in the Service Booking Report, the Repair Pipeline, the Workshop Calendar and the dashboard counts.
Related FR	FR-01, FR-02, FR-03, FR-05

Table 3.6: Use Case Description – Update Repair Status

Field	Description
Use Case ID	UC-02
Use Case Name	Update Repair Status
Actor(s)	Staff / Admin
Pre-condition	A service booking record exists.
Main Flow	1. Staff opens the Repair Pipeline or Service Booking Report. 2. Staff selects the booking and edits the Status field (or moves the card to another column). 3. The record is updated. 4. The pipeline columns and dashboard counts reflect the new status.
Alternate Flow	If a spare part is not available, the status is set to “Awaiting Parts” until the part arrives.
Post-condition	The watch is shown under its new status; when set to “Ready for Pickup” it is counted in the “Ready to collect” tile.
Related FR	FR-04, FR-07, FR-09

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CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

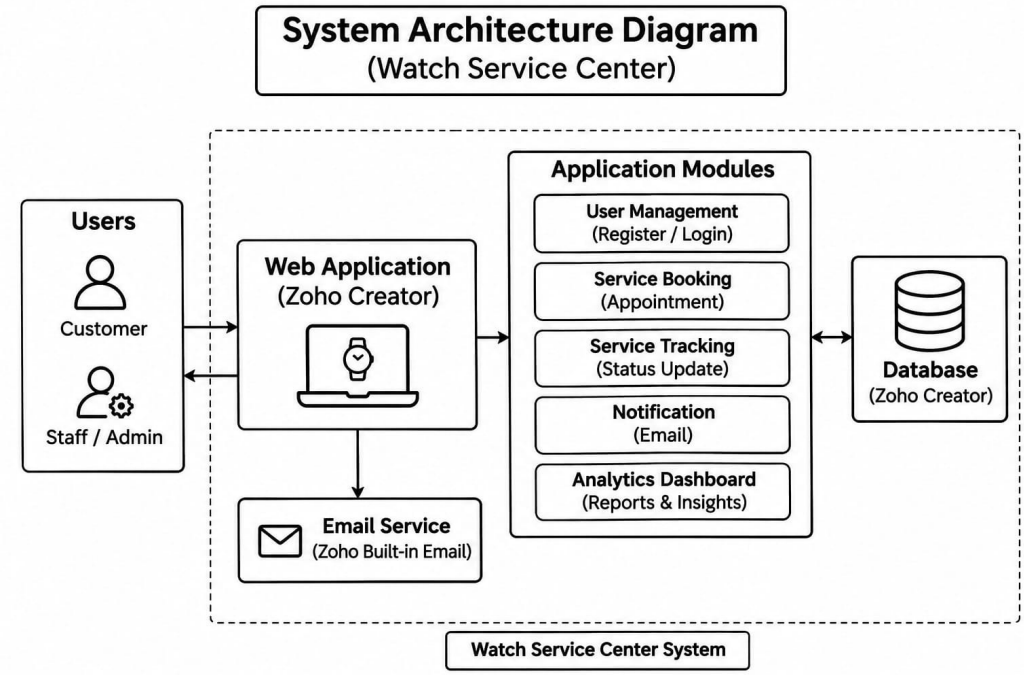


Figure 4.1: System Architecture of the Watch Service Center

Customers and staff interact with a web application built on Zoho Creator. The application is organised into five modules — user management, service booking, service tracking, notification and analytics dashboard — which read from and write to the Zoho Creator database. Notifications are sent through the e-mail service built into the Zoho platform, so no separate mail server is required.

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4.2 Data Model

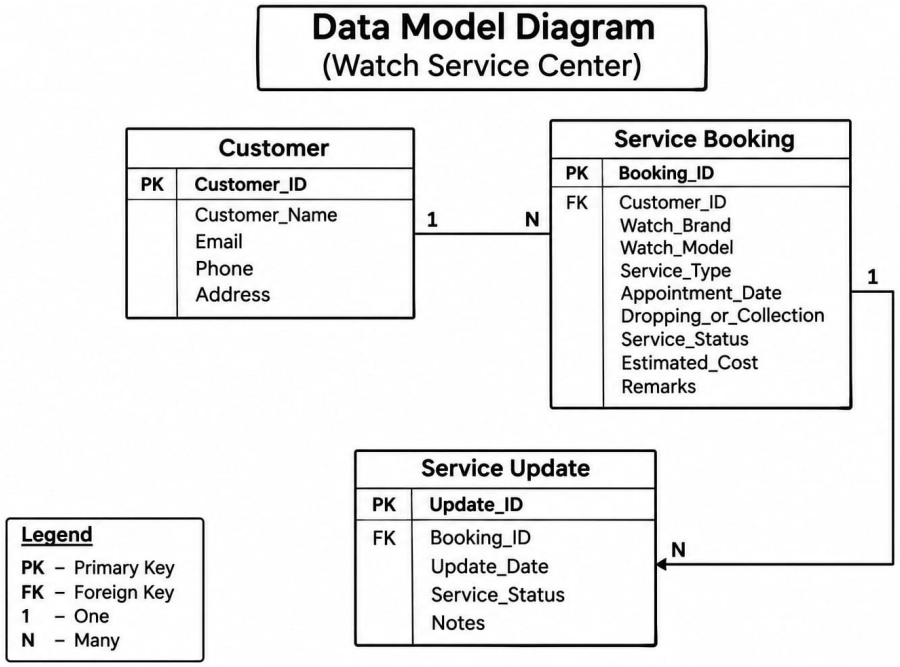


Figure 4.2: Data Model Diagram of the Watch Service Center

The logical data model consists of three entities: Customer, Service Booking and Service Update. One customer can make many service bookings, and one booking can have many service updates. In the current Zoho Creator implementation these entities are consolidated into a single Service_Booking form: the customer attributes are stored directly on the booking record, and the service update trail is represented by the Status field that moves along the repair pipeline. This keeps data entry to a single screen for the counter staff. Separating Customer and Service Update into their own forms (with a lookup and a subform) is noted as a future enhancement.

Table 4.1: Entities and Relationships

Entity (Logical)	Key Fields	Related To	Relationship Type	Implemented Using
Customer	Customer_Name, Email, Phone, Address	Service Booking	One-to-Many	Customer fields stored on the Service_Booking form
Service Booking	Watch brand and model, issue, serial number, date, time, status	Customer, Service Update	Many-to-One / One-to-Many	Service_Booking form (primary form)
Service Update	Update date, service status, notes	Service Booking	Many-to-One	Status picklist on Service_Booking, shown in the Repair Pipeline

4.3 Form Design

Table 4.2: Field Specification – Service Booking Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Customer Name	Customer_Name	Name	Yes	First / Last name; prefix and suffix hidden	Personal data
Customer Email	Customer_Email	Email	Yes	Valid e-mail format; max 80 characters	Personal data
Customer Address	Customer_Address	Address	No	Address lines, city, state, postal code, country	Captures map coordinates; personal data
Customer Phone	Customer_Phone	Phone Number	Yes	Country code selector (default +91)	Personal data
Watch brand and model	Watch_brand_and_model	Single Line	No	—	Used in the e-mail subject and calendar
Issue or replacement description	Issue_or_replacement	Multi Line	No	Height 100 px	Describes the fault or part to replace
Serial Number	Serial_Number	Number	No	Numeric only	Identifies the watch
Dropping or Collecting Date	Dropping_or_Collecting_Date	Date	No	All days of the week allowed	Drives the Workshop Calendar
Dropping or Collecting Time	Dropping_or_Collecting_Time	Time	No	Format hh:mm:ss	Appointment time
Status	Status	Drop Down (picklist)	No	Checked In, Under Diagnosis, Awaiting Parts, On the Bench, Ready for Pickup	Drives the pipeline and dashboard

The form shows “Data Added Successfully!” as its success message. On add, the form displays a Submit button; on edit, it displays Update and Cancel buttons. The complete form definition is given in Appendix C.

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4.4 Process and Workflow Design

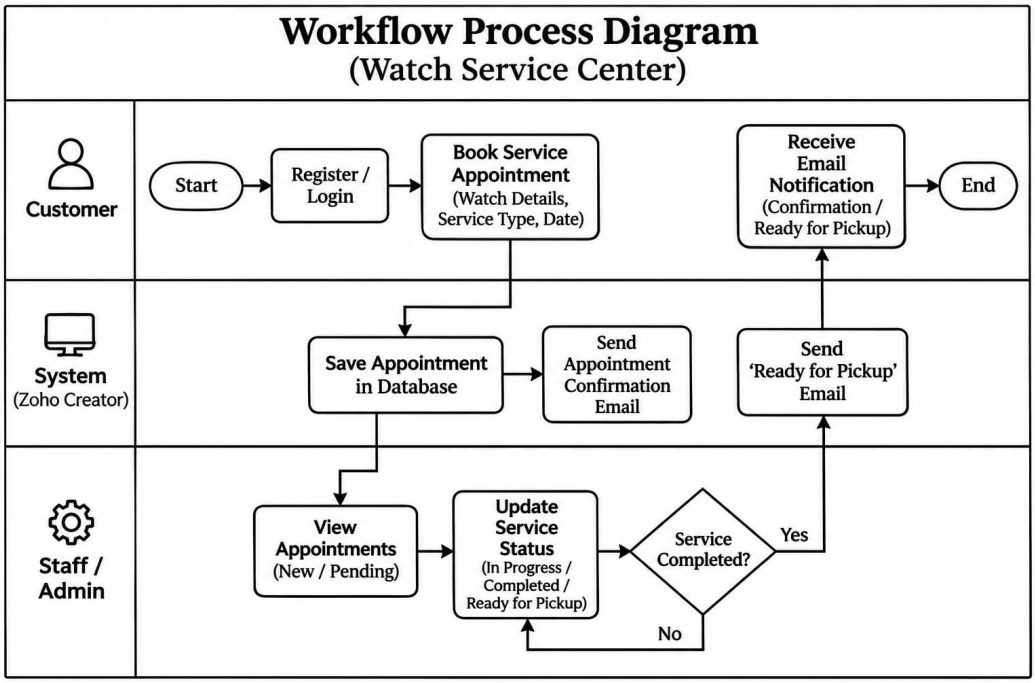


Figure 4.3: Workflow Process Diagram of the Watch Service Center

The swim-lane diagram shows how work flows between the customer, the system (Zoho Creator) and the staff. The customer books a service appointment; the system saves it in the database and sends a confirmation e-mail; staff view new and pending appointments and update the service status until the service is completed; and a "Ready for Pickup" e-mail notifies the customer. In this release the booking, database save, confirmation e-mail, appointment viewing and status updating steps are implemented. The "Ready for Pickup" e-mail and customer login steps are part of the design and are listed under future enhancements.

4.5 User Interface and Navigation Design

The application uses a dark left navigation pane. The Service Booking section expands to show four components and the Analytics Dashboard page sits below it.

Table 4.3: Navigation Structure of the Application

Menu Item	Component Type	Source / Grouping	Purpose
Service Booking	Form	Service Booking	Capture a new service booking
Service Booking Report	List report	Service Booking	Tabular list of all bookings with contact, address and watch details
Repair Pipeline	Kanban report	Grouped by Status	Visual board of watches in each repair stage
Workshop Calendar	Calendar report	Dropping_or_Collecting_ Date	Month / week / day view of appointments

Menu Item	Component Type	Source / Grouping	Purpose
Analytics Dashboard	Page	Service Booking	KPI tiles and status distribution chart

4.6 Access Control Design

The design defines two operational roles in addition to the application owner. In the present build the application is administered by the owner account; the matrix below documents the permissions to be assigned when additional staff and customers are onboarded.

Table 4.4: Role–Permission Matrix (A – Add, E – Edit, V – View, D – Delete)

Component	Administrator	Staff	Customer (planned)
Form: Service Booking	A E V D	A E V	A
Report: Service Booking Report	V	V	—
Report: Repair Pipeline	V (move cards)	V (move cards)	—
Report: Workshop Calendar	V	V	—
Page: Analytics Dashboard	V	V	—

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CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “Watch Service Center” (link name watch-service-center) was created in Zoho Creator. The left navigation pane groups the working components under a Service Booking section — the Service Booking form, Service Booking Report, Repair Pipeline and Workshop Calendar — followed by the Analytics Dashboard page. The application is published at https://creatorapp.zoho.in/tharunkrishna.2024.csbs_r/watch-service-center.

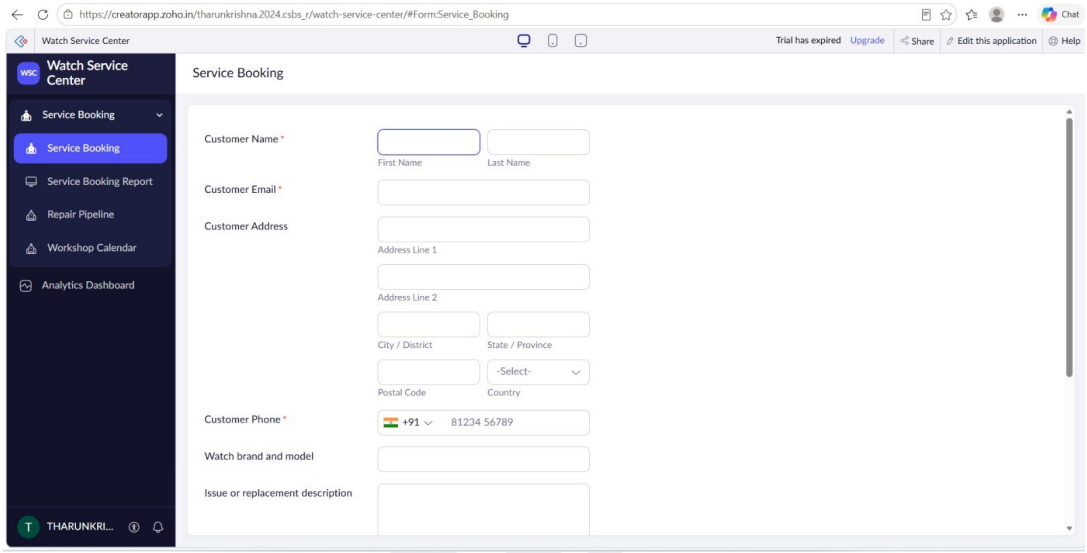


Figure 5.1: Application Navigation and the Service Booking Form in Live Mode

5.2 Forms and Data Capture

A single Service Booking form (Service_Booking) was built. It uses ten different field types — name, e-mail, address, phone number, single-line text, multi-line text, number, date, time and drop-down — as specified in Table 4.2. Customer name, e-mail and phone number are mandatory; the name, e-mail, address and phone fields are marked as personal data. The address field is configured to capture map coordinates, and the phone field shows a country-code selector defaulting to +91. The Status field is a drop-down holding the five repair stages, which are later used to group the Kanban pipeline and the dashboard chart.

Code 5.1 shows how the Status picklist is declared in the form definition.

```
Status
(
    type = picklist
    maxchar = 100
    values = {"Checked In","Under Diagnosis","Awaiting Parts",
              "On the Bench","Ready for Pickup"}
    row = 1
    column = 1
    width = medium
)
```

Code 5.1: Status Field Definition (Service_Booking form)

Seven sample service records were entered to test and demonstrate the application (Table 5.1).

Table 5.1: Sample Service Records Used for Testing

S.No.	Customer	Watch Brand and Model	Status
1	Shantanu Kumar	Aigner Silverstar	On the Bench
2	Reddy Gaaru	Cartier Galaxy	Ready for Pickup
3	Vijay Guru	Hublot trackcrew	On the Bench
4	Jithu Sharma	Citizen Marine diver	Awaiting Parts
5	Sai Kiran	Seiko Polaroid	Under Diagnosis
6	Vichu Nath	Tag Heuer Celestic	Ready for Pickup
7	Sudhir Gupta	Omega Seamaster	Checked In

All customer details in the sample records are test data.

5.3 Reports

Table 5.2: Reports Implemented

Report Name	Type	Source Form	Grouping / Sorting	Custom Actions	Roles
Service Booking Report	List / Grid	Service Booking	Sortable and groupable columns; search	Add record (+) button; record edit	Administrator, Staff
Repair Pipeline	Kanban	Service Booking	Grouped by Status (five columns)	Drag a card to change its stage	Administrator, Staff
Workshop Calendar	Calendar	Service Booking	By Dropping_or_Collecting_Date; Month / Week / Day views	Add record; Today navigation	Administrator, Staff
Status Distribution	Pie chart (on dashboard)	Service Booking	Grouped by Status	Filter icon on the chart	Administrator, Staff

The Service Booking Report lists customer name, e-mail, address, phone and watch details for every booking, with a search icon and an add-record button in the title bar.

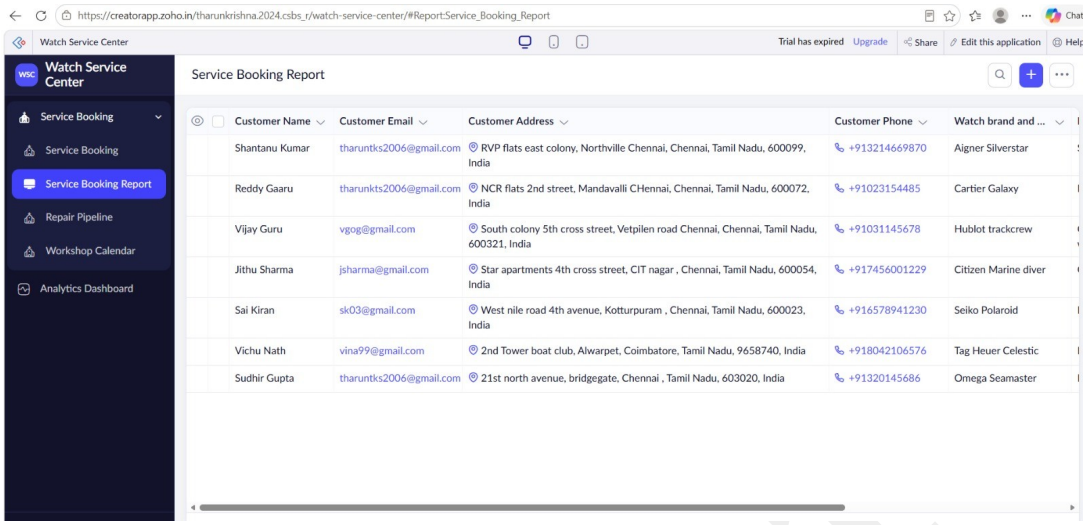


Figure 5.2: Service Booking Report (List Report)

Each of the seven sample bookings is one row; columns can be sorted, grouped and searched, and the + button adds a new booking.

The Repair Pipeline is a Kanban report grouped by the Status field. Each card shows the customer name and address, and the five columns — Checked In, Under Diagnosis, Awaiting Parts, On the Bench and Ready for Pickup — give the workshop an instant picture of where every watch stands.

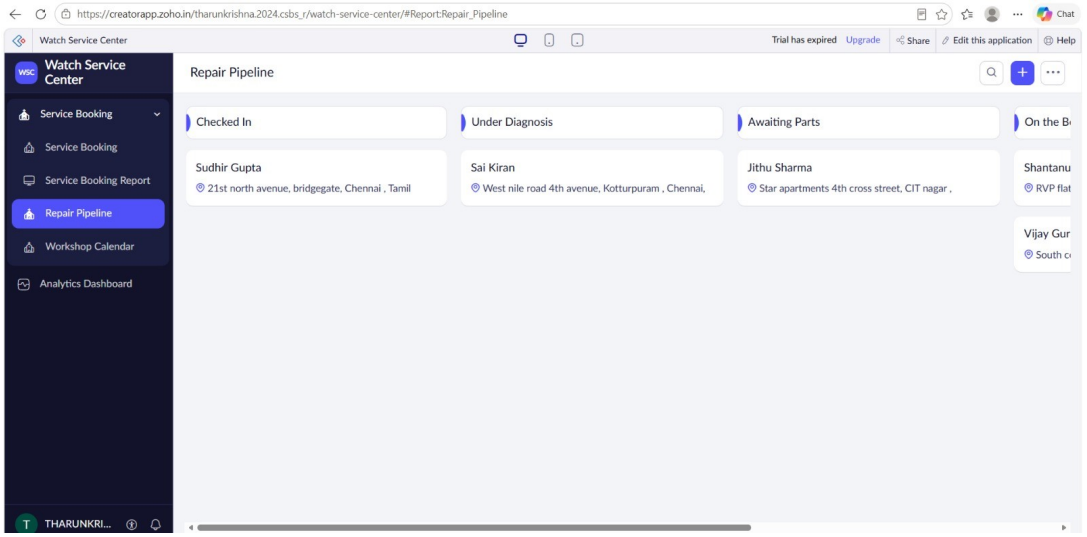


Figure 5.3: Repair Pipeline (Kanban Report)

Sudhir Gupta is Checked In, Sai Kiran is Under Diagnosis, Jithu Sharma is Awaiting Parts and two watches are On the Bench; scroll right for Ready for Pickup.

The Workshop Calendar plots each booking against its dropping or collecting date, using the watch brand and model as the event label. In the November 2026 view, the Aigner Silverstar and Cartier Galaxy bookings appear on 1 November.

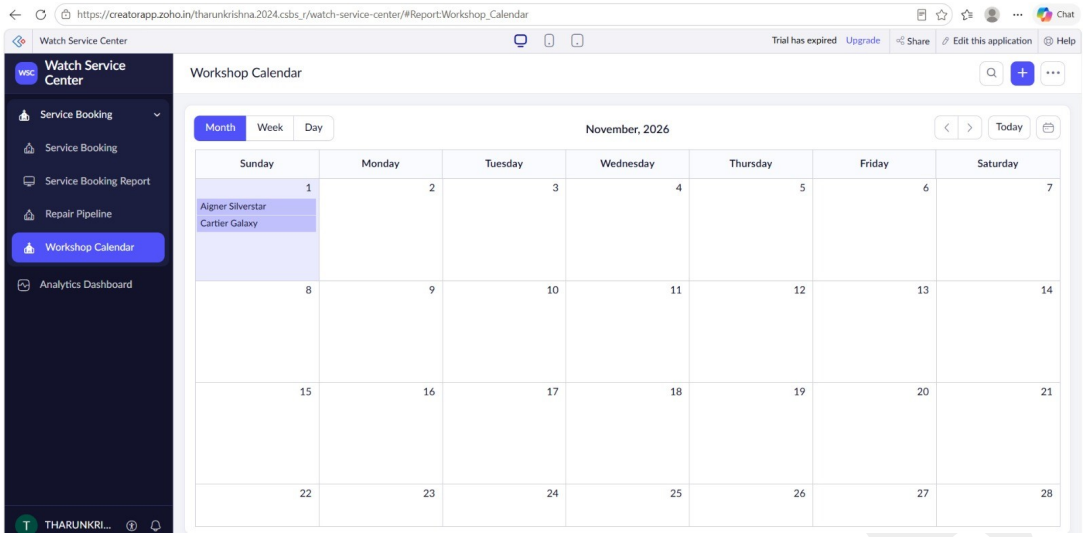


Figure 5.4: Workshop Calendar (Calendar Report)

The calendar offers Month, Week and Day views, a Today shortcut and arrows to move between months. Each event carries the watch brand and model, so the workshop can see at a glance which watches are due for dropping or collecting on a given day.

5.4 Pages and Dashboards

The Analytics Dashboard was built as a Zoho Creator page. It contains a pie chart showing the distribution of units by status (legend: Checked In, Under Diagnosis, Awaiting Parts, On the Bench and Ready for Pickup), a “Total units” tile (7 for the sample data), a “Ready to collect” tile (2) and an additional summary tile. The chart has a filter control so that the view can be narrowed.

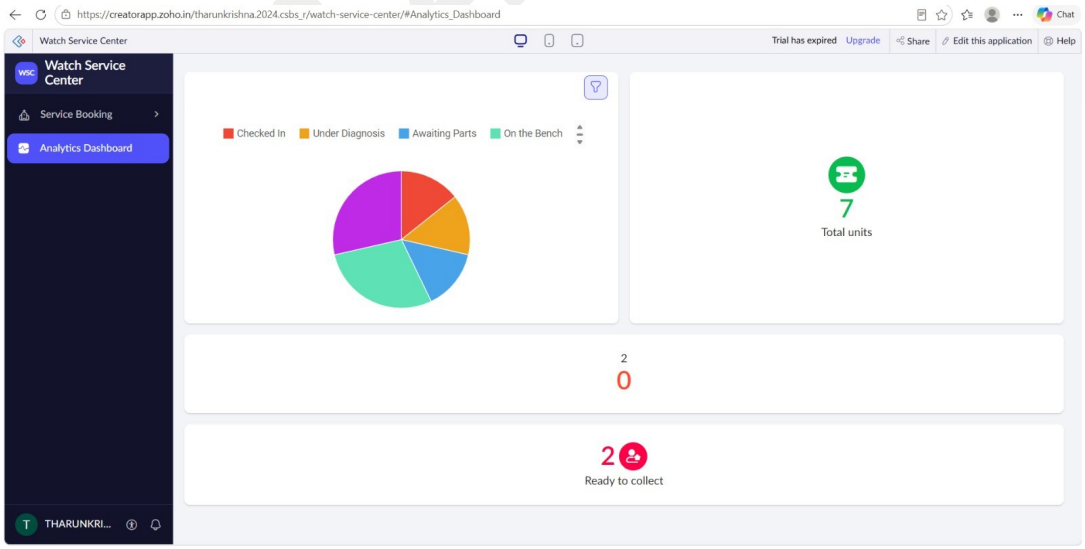


Figure 5.5: Analytics Dashboard

The pie chart splits the seven units across the five repair stages, while the Total units tile shows 7 and the Ready to collect tile shows 2, matching the two Ready for Pickup records in Table 5.1. The filter icon on the chart narrows the view.

5.5 Deluge Scripting

Table 5.3: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Service Booking → Created → Successful form submission (workflow “Send Welcome Email”)	Send a confirmation e-mail that quotes the customer name and the watch brand and model	sendmail task, input variables, string concatenation, HTML in message body, zoho.adminuserid

The script builds the subject line by joining a fixed text with input.Watch_brand_and_model, and builds an HTML message that greets the customer using input.Customer_Name and repeats the watch details. During development the “to” address was kept as the application administrator’s own address so that the developer could verify the mail; the planned change to input.Customer_Email is recorded in the defect log (Table 6.2).

```
// Script ID : DS-01
// Trigger   : Service Booking -> Created -> Successful form submission
// Purpose   : Send a confirmation e-mail after a booking is saved
sendmail
[
    from :zoho.adminuserid
    to :zoho.adminuserid
    subject :"Watch Service Appointment Confirmation - "
        + input.Watch_brand_and_model
    message :"Hi " + input.Customer_Name + ",<br><br>"
        + "Your watch service appointment has been "
        + "successfully registered.<br><br>"
        + "<b>Watch Details:</b><br>"
        + "Watch: " + input.Watch_brand_and_model + "<br><br>"
        + "Thank you for choosing our Watch Service Center.<br>"
        + "We will contact you regarding your appointment."
]
```

Code 5.2: Deluge Script for the Confirmation E-mail (DS-01)

5.6 Workflows and Automation

Table 5.4: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Send Welcome Email	Form workflow	Service Booking → Created → Successful form submission	Execute without condition	Deluge script sends confirmation e-mail	DS-01

The workflow is configured under the Service Booking form so that it runs every time a new record is successfully added. It was created with the workflow builder and its single action is the Deluge script in Code 5.2.

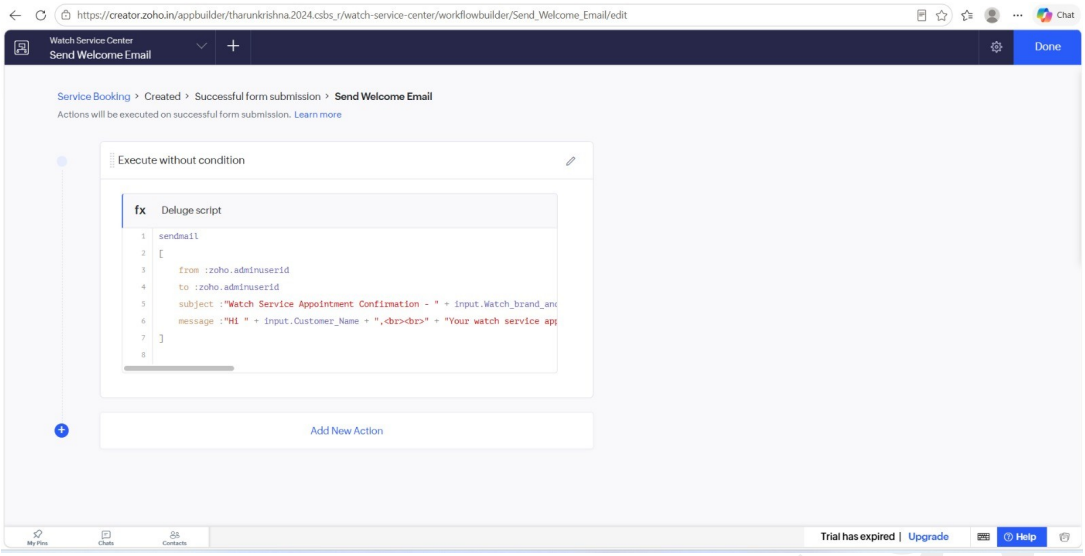


Figure 5.6: Workflow Builder – "Send Welcome Email" Form Workflow

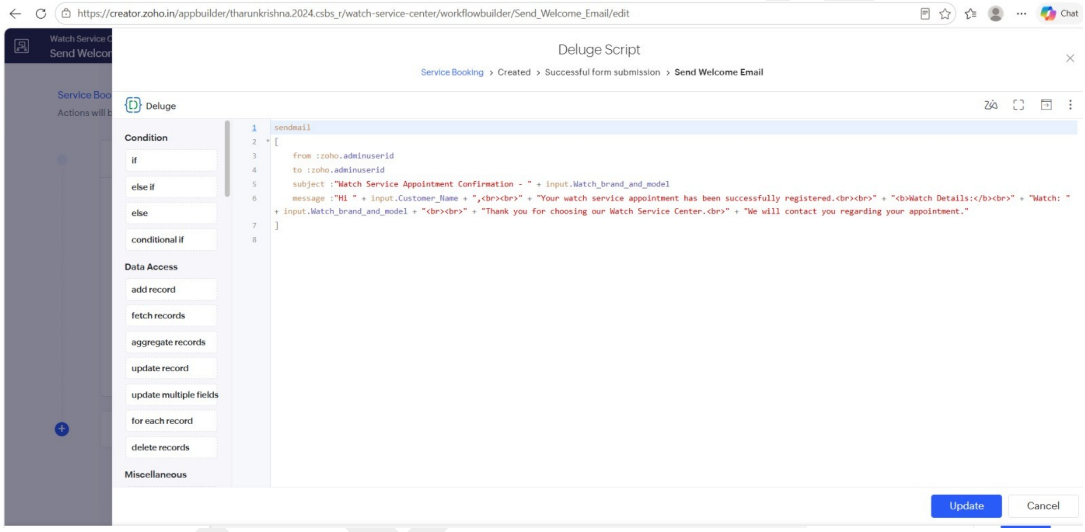


Figure 5.7: Deluge Script Editor with the sendmail Script

5.7 Integrations

Table 5.5: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho Mail	Zoho mail service built into Zoho Creator	sendmail Deluge task	Zoho system authentication (zoho.adminuserid)	Dispatch the booking confirmation e-mail

The confirmation e-mail sent for the Aigner Silverstar booking was received in the Gmail mobile app at 7:03 pm, with the subject "Watch Service Appointment Confirmation – Aigner Silverstar", a greeting to the customer and the watch details. No external REST API integration was required in this release.

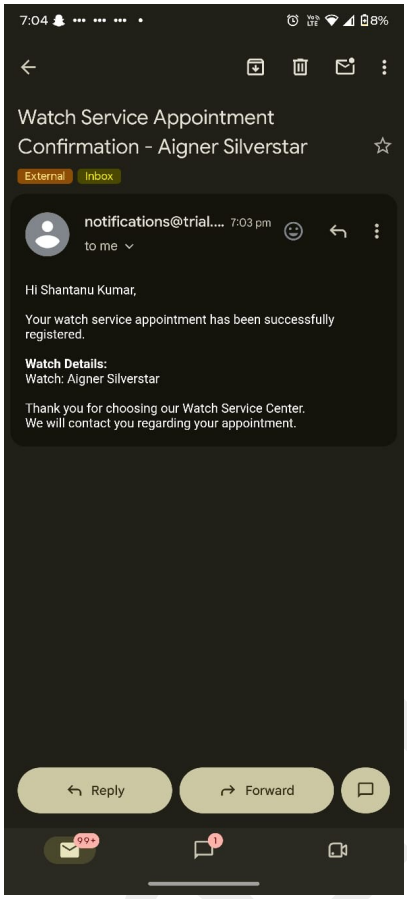


Figure 5.8: Booking Confirmation E-mail Received on a Mobile Device

5.8 User Management and Role-Based Access

Access to the application is controlled through Zoho authentication, and the application is administered by its owner account. The name, e-mail, address and phone fields are flagged as personal data in the form definition. Staff and customer profiles are designed in Table 4.4 and can be assigned to additional users as the service centre grows; a customer-facing portal with record-level criteria such as “Added User == zoho.loginuser” is part of the future scope.

5.9 AI and Analytics Features

Analytics are provided through the dashboard page: a pie chart of units by status and KPI tiles for total units and units ready to collect. Zia AI features (for example, predictive turnaround estimates or automatic categorisation of the issue description) were not enabled in this release and are noted as a future enhancement.

5.10 Pro-Code Extension

No pro-code extension was needed for the current scope. If SMS or WhatsApp notifications are added later, they can be implemented through a custom function, a REST connection or a Zoho Catalyst serverless function called from the same form workflow.

5.11 Mobile and Portal Access

Zoho Creator renders the application responsively; the top bar offers desktop, tablet and mobile device previews, and the Zoho Creator mobile app can be used to open the same application. The confirmation e-mail was verified on a mobile device (Figure 5.8). A dedicated customer portal is a planned enhancement.

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CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out at three levels: functional testing of the Service Booking form (data entry and mandatory-field validation); workflow testing of the Deluge confirmation e-mail by submitting a booking and checking the received message; and verification of the reports and dashboard against the seven sample records. A scenario-based user acceptance walk-through was then performed (Section 6.5).

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01, FR-03	Submit a complete service booking	Fill all fields and click Submit	Record saved; success message shown	Record saved; seven sample records present	Pass
TC-02	FR-02	Submit without a mandatory field	Leave e-mail blank and click Submit	Submission blocked; field flagged	Submission blocked	Pass
TC-03	FR-05	Confirmation e-mail on booking	Submit the Aigner Silverstar booking	E-mail with watch name in subject and body	E-mail received at 7:03 pm (Fig. 5.8)	Pass
TC-04	FR-06	Open the Service Booking Report	Click Service Booking Report	All bookings listed with address, phone and watch	Seven records listed	Pass
TC-05	FR-04, FR-07	View the Repair Pipeline	Click Repair Pipeline	Cards grouped under status columns	Cards shown under their statuses	Pass
TC-06	FR-08	View the Workshop Calendar	Open November 2026 month view	Bookings plotted on their dates	Two bookings on 1 Nov 2026	Pass
TC-07	FR-09	Check dashboard figures	Open Analytics Dashboard	Total units = record count; Ready to collect = Ready for Pickup records	Total 7; Ready to collect 2	Pass

6.3 Debugging Deluge Scripts

The e-mail script was checked with the Deluge editor’s syntax validation and by repeated test submissions of the form. The subject and body of the received message were compared with the

expected text to confirm that input.Watch_brand_and_model and input.Customer_Name were substituted correctly. Further statements such as info can be added to the script to log values during troubleshooting.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied / Planned	Status
BUG-01	Confirmation e-mail is delivered to the administrator address instead of the customer	“to” in the sendmail task was set to zoho.adminuserid for developer testing	Change “to” to input.Customer_Email once the sender is configured for external recipients	Open (planned)
BUG-02	A seven-digit postal code (9658740) was accepted for the Coimbatore booking	No format validation on the postal code sub-field	Add an On Validate check for a six-digit PIN code	Open (planned)

6.5 User Acceptance Testing

Table 6.3: User Acceptance Scenarios

Scenario	Role Tested	Observation	Action Taken / Planned
Register a watch for service at the counter	Staff	Form is simple; confirmation e-mail received immediately after submission	Route e-mail to the customer address (BUG-01)
Check where each watch is in the workshop	Staff	Kanban columns show every watch by stage; the fifth column needs horizontal scrolling on a narrow screen	Accepted; column widths can be tuned later
Check appointments for the day	Staff	Calendar shows the watches due on 1 November 2026	No change required
Review workload	Administrator	Dashboard shows 7 total units and 2 ready to collect, with the status pie chart	Add more KPI tiles (for example, units awaiting parts)

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed in the Zoho Creator builder (creator.zoho.in) and accessed by users in live mode (creatorapp.zoho.in), which is the URL used for all report and dashboard screenshots in this report. Changes to the form, workflow and reports were made in the builder and checked in live mode before the final demonstration.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases in Table 6.1 executed	✓
2	Sample data clearly identified as test data	✓
3	Form workflow enabled and verified with a test e-mail	✓
4	Reports and dashboard verified in live mode	✓
5	Application URL verified	✓
6	Staff and customer profiles created for additional users	Planned
7	Customer recipient address configured in the e-mail script	Planned

7.2 Performance Optimisation

For a single-branch service centre the data volume is small, so performance relied mainly on keeping the design simple. No timing benchmarks were recorded for this release.

Table 7.2: Performance Considerations

Design Choice	Component	Benefit
Single consolidated form	Service Booking	No lookups or joins; every report reads one form
Reports driven directly by form fields	Kanban, Calendar, Dashboard	Views refresh instantly when the Status or date changes
One lightweight Deluge script	DS-01	Only a single sendmail task runs on submission; no loops or external calls

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho account authentication	Application access	Unauthorised access to booking data
Personal-data flag on fields	Customer name, e-mail, address, phone	Mishandling of customer personal information

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Security Control	Where Applied	Threat Mitigated
No credentials in scripts (zoho.adminuserid used for sender)	DS-01	Leakage of passwords or API keys
Mandatory-field and field-type validation	Service Booking form	Incomplete or malformed data entry

7.4 Governance and Administration

The application owner acts as administrator, controlling the form, reports, workflow and page definitions. Zoho Creator maintains audit information about application changes and user activity for administrators, and the same facility can be used to review booking changes as more staff are added. Documented defects (Table 6.2) and planned items (Table 7.1) provide a simple change log for the next release.

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CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The operational scenario below follows a single watch through the application, using the figures presented in Chapter 5 as evidence.

Table 8.1: Operational Scenario – Aigner Silverstar Booking

Step	Action	Result / Evidence
1	Staff open the Service Booking form and enter the customer and watch details	Form with all fields (Figure 5.1)
2	Staff submit the form	Record saved and the “Send Welcome Email” workflow runs (Figures 5.6, 5.7)
3	Customer receives the confirmation	E-mail with the watch name and a thank-you message (Figure 5.8)
4	Booking appears in the reports	Row in the Service Booking Report and card in the Repair Pipeline (Figures 5.2, 5.3)
5	Appointment is visible by date	Entry on 1 November 2026 in the Workshop Calendar (Figure 5.4)
6	Management reviews the workload	Dashboard shows total units, ready-to-collect count and the status chart (Figure 5.5)

8.2 Achievement of Objectives

Table 8.2: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Capture booking details with validation	Sections 4.3, 5.2	Service Booking form with ten field types and mandatory fields (Figure 5.1)
Track watches through five repair stages	Section 5.3	Repair Pipeline Kanban report (Figure 5.3)
Automate confirmation e-mail	Sections 5.5, 5.6	Workflow and Deluge script; e-mail received (Figure 5.8)
Provide calendar and dashboard	Sections 5.3, 5.4	Workshop Calendar and Analytics Dashboard (Figures 5.4, 5.5)
Build and publish on Zoho Creator	Chapter 7	Application accessible at the live URL

8.3 Comparison with the Existing System

Table 8.3: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Booking record	Handwritten register	Digital form with mandatory fields	Complete, searchable records
Customer acknowledgement	None or verbal	Automatic confirmation e-mail	Written confirmation at check-in
Repair status tracking	Verbal enquiry to the workshop	Five-stage Kanban pipeline	Live view of every watch
Appointment scheduling	Paper diary	Workshop calendar	Day / week / month visibility
Workload overview	Manual counting	Analytics dashboard	Instant totals and status split

8.4 Limitations of the Low-Code Approach

Some limitations were met during the project. The dashboard offers a fixed set of widgets and limited control over detailed styling. The Kanban board shows only a few fields on each card, and its fifth column requires horizontal scrolling on smaller screens. Outbound e-mail from the sendmail task is restricted to configured sender addresses, which is why the recipient was kept as the administrator during testing. The application was built on the trial plan, so continued development after the trial period would require a paid plan.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The form, the Kanban pipeline, the calendar and the dashboard were created entirely with no-code configuration. The only low-code element was the Deluge script for the confirmation e-mail. Features such as SMS or WhatsApp messaging, payment collection or advanced analytics would move the solution toward the pro-code end of the spectrum, using REST connections or a Catalyst function.

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CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Watch Service Center application replaces paper registers and diaries with a single digital system. Counter staff record a booking once, the customer receives an automatic confirmation e-mail, the workshop sees each watch on a Kanban board, appointments appear in a calendar and management monitors the workload on a dashboard. The project demonstrates that a complete service-management solution can be delivered quickly on Zoho Creator using forms, reports, a page, a form workflow and one Deluge script.

9.2 Future Enhancements

- Send the confirmation e-mail to the customer's own address (input.Customer_Email) and add the "Ready for Pickup" notification shown in the workflow design.
- Add a customer portal with login so that customers can track the status of their own watch.
- Split the data model into separate Customer and Service Update forms using a lookup and a subform to keep a full status history.
- Add service type, estimated cost and remarks fields, followed by quotation and invoicing.
- Add postal code and serial number validation, and SMS / WhatsApp notifications.
- Create Staff and Customer role profiles with record-level access criteria.

9.3 Learning Outcomes and Career Relevance

Developing this project provided hands-on experience in designing forms, building Kanban, calendar and list reports, creating a dashboard page, writing Deluge scripts and configuring form workflows. These skills are relevant to roles in low-code development, business-process automation and solution consulting.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist (Y – Implemented, Partial – Partly implemented, N – Not implemented)

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections / menu organised	Y	Service Booking section with four components; Analytics Dashboard	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	Y	Service Booking form (10 field types)	4.3, 5.2	Table 4.2, Figure 5.1
II	Lookup field (relationship between forms)	N	Single consolidated form; proposed as future work	4.2, 9.2	—
II	Subform (parent–child data)	N	Not used in this release	4.2, 9.2	—
II	Field validations / layout rules	Y	Mandatory name, e-mail, phone; e-mail / phone / number field types	4.3, 5.2	Table 4.2
II	List report with filter, grouping, custom action	Partial	Service Booking Report (sort, search, add record)	5.3	Figure 5.2
II	Kanban / Calendar / Timeline / Map report	Y	Repair Pipeline (Kanban) and Workshop Calendar	5.3	Figures 5.3, 5.4
II	Pivot table / chart report	Y	Status pie chart on the dashboard	5.4	Figure 5.5
II	Page / dashboard built with UI Builder	Y	Analytics Dashboard page	5.4	Figure 5.5
II	Deluge – form events (validate / success / field actions)	Y	Successful form submission workflow (DS-01)	5.5	Code 5.2
II	Deluge – functions, fetch, loops, collections	N	Not required for the current scope	—	—
II	Email / in-app notifications	Y	Booking confirmation e-mail	5.5, 5.6	Figure 5.8
II	Form workflow	Y	Send Welcome Email	5.6	Table 5.4, Figure 5.6
II	Scheduled workflow	N	Not implemented	—	—
II	Approval process	N	Not implemented	—	—
II	Blueprint	N	Status stages handled through picklist and Kanban	—	—
II	Batch / report workflow	N	Not implemented	—	—
II	Testing & debugging evidence	Y	Test cases and received e-mail	6	Table 6.1, Figure 5.8

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Integration with a Zoho service	Y	Zoho Mail through the sendmail task	5.7	Table 5.5
III	Connection / REST API call (invokeurl)	N	Not required in this release	—	—
III	Webhook or Creator REST API	N	Not implemented	—	—
III	Users, roles and permission profiles	Partial	Owner administers; roles designed in the permission matrix	4.6, 5.8	Table 4.4
III	Record-level access criteria / customer portal	N	Planned enhancement	5.8, 9.2	—
III	Zia / AI feature	N	Planned enhancement	5.9, 9.2	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard KPIs and pie chart	5.4	Figure 5.5
IV	Performance optimisation applied & measured	N	Design considerations only; no benchmark recorded	7.2	Table 7.2
IV	Deployment across environments	Partial	Builder and live-mode access	7.1	Table 7.1
IV	Security controls (encryption, OAuth, permissions)	Partial	Zoho authentication, personal-data flags, no credentials in script	7.3	Table 7.3
IV	Governance (audit trail, administration)	Partial	Owner administration; change log through defect and checklist tables	7.4	Table 6.2
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Proposed for SMS / WhatsApp notifications	5.10	—
V	Low-code limitations & spectrum analysis	Y	Discussed in results chapter	8.4, 8.5	—
—	Mobile app / portal access	Partial	Responsive layout and device preview; e-mail verified on mobile	5.11	Figure 5.8

Total features fully implemented: 13 / 32; partially implemented: 6 / 32

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APPENDIX B – COURSE OUTCOME MAPPING

Guidance: Show where each Course Outcome of CB23F35 is evidenced in this report and application. Self-assess the attainment level (1 – Low, 2 – Medium, 3 – High); the examiner may revise it.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3 (High)	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3 (High)	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2 (Medium)	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2 (Medium)	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2 (Medium)	

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APPENDIX C – COMPLETE DELUGE SOURCE CODE

Guidance: Every script of the application is given in full, grouped by form / workflow, with a header comment giving the Script ID, location / trigger and purpose.

C.1 Service Booking Form Definition

Form definition (.ds) of the Service_Booking form, including its fields, success message and actions.

```
form Service_Booking
{
    displayname = "Service Booking"
    success message = "Data Added Successfully!"
    Section
    (
        type = section
         row = 1
         column = 0
        width = medium
    )
    must have Customer_Name
    (
        type = name
        displayname = "Customer Name"
         prefix
         (
              type = prefix
              displayname ="Prefix"
              visibility = false
              value = {"Mr.","Mrs.","Ms."}
         )
         first_name
         (
              type = first_name
              displayname ="First Name"
         )
         last_name
         (
              type = last_name
              displayname ="Last Name"
         )
         suffix
         (
              type = suffix
              displayname ="Suffix"
              visibility = false
         )
         row = 1
         column = 1
        width = medium
        personal data = true
    )
    must have Customer_Email
    (
```

```
        type = email
        displayname = "Customer Email"
        maxchar = 80
         row = 1
         column = 1
        width = medium
        personal data = true
    )
    Customer_Address
    (
        type = address
        displayname = "Customer Address"
         capture_coordinates = true
         adjust_using_map = false
         address_line_1
         (
               type = address_line_1
               displayname = "Address Line 1"
         )
         address_line_2
         (
               type = address_line_2
               displayname = "Address Line 2"
         )
         district_city
         (
               type = district_city
               displayname = "City / District"
         )
         state_province
         (
               type = state_province
               displayname = "State / Province"
         )
         postal_Code
         (
               type = postal_Code
               displayname = "Postal Code"
         )
         country
         (
               type = country
               displayname = "Country"
         )
         latitude
         (
               type = latitude
               displayname = "latitude"
              visibility = false
         )
         longitude
         (
               type = longitude
               displayname = "longitude"
              visibility = false
         )
         row = 1
```

```
     column = 1
    width = medium
    personal data = true
)
must have Customer_Phone
(
    type = phonenumber
    displayname = "Customer Phone"
     row = 1
     column = 1
    width = medium
    personal data = true
)
Watch_brand_and_model
(
    type = text
    displayname = "Watch brand and model"
     row = 1
     column = 1
    width = medium
)
Issue_or_replacement
(
    type = textarea
    displayname = "Issue or replacement description"
    height = 100px
     row = 1
     column = 1
    width = medium
)
Serial_Number
(
    type = number
    displayname = "Serial Number"
     row = 1
     column = 1
    width = medium
)
Dropping_or_Collecting_Date
(
    type = date
    displayname = "Dropping or Collecting Date"
    alloweddays = 0,1,2,3,4,5,6
     row = 1
     column = 1
    width = medium
)
Dropping_or_Collecting_Time
(
    type = time
    displayname = "Dropping or Collecting Time"
    timedisplayoptions = "hh:mm:ss"
     row = 1
     column = 1
    width = medium
)
Status
```

```
    (
        type = picklist
        maxchar = 100
        values = {"Checked In","Under Diagnosis","Awaiting Parts","On the Bench","Ready for
Pickup"}
         row = 1
         column = 1
        width = medium
    )

    actions
    {
        on add
        {
            submit
            (
                   type = submit
                   displayname = "Submit"
            )
        }
        on edit
        {
            update
            (
                   type = submit
                   displayname = "Update"
            )
            cancel
            (
                   type = cancel
                   displayname = "Cancel"
            )
        }
    }
}
```

C.2 Send Welcome Email – Form Workflow Script

Script DS-01, executed on successful form submission (line breaks added for readability).

```
// Script ID : DS-01
// Trigger   : Service Booking -> Created -> Successful form submission
// Purpose   : Send a confirmation e-mail after a booking is saved
sendmail
[
    from :zoho.adminuserid
    to :zoho.adminuserid
    subject :"Watch Service Appointment Confirmation - "
        + input.Watch_brand_and_model
    message :"Hi " + input.Customer_Name + ",<br><br>"
        + "Your watch service appointment has been "
        + "successfully registered.<br><br>"
        + "<b>Watch Details:</b><br>"
        + "Watch: " + input.Watch_brand_and_model + "<br><br>"
        + "Thank you for choosing our Watch Service Center.<br>"
        + "We will contact you regarding your appointment."
]
```

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APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the production URL https://creatorapp.zoho.in/tharunkrishna.2024.csbs_r/watch-service-center in any web browser, or open the Zoho Creator mobile app and select Watch Service Center from the workspace. Sign in with the Zoho account that has been given access to the application (shared by the application owner). The left navigation pane shows the Service Booking section and the Analytics Dashboard.

D.2 Instructions for Staff / Admin

- **Register a service booking:** Click Service Booking in the left pane. Enter the customer's first and last name, e-mail, address, phone number, watch brand and model, issue description, serial number and the dropping or collecting date and time. Choose the current Status (normally Checked In) and click Submit. A success message appears and a confirmation e-mail is sent.
- **View all bookings:** Open Service Booking Report to see every record with contact and watch details. Use the search icon to find a customer, or the + button to add a new booking.
- **Track and update repair status:** Open Repair Pipeline. Each card is one watch, placed in the column of its current status. Move the card, or edit the record and choose a new Status, as the watch moves from Checked In to Under Diagnosis, Awaiting Parts, On the Bench and Ready for Pickup.
- **Check appointments:** Open Workshop Calendar and switch between Month, Week and Day views to see the watches due for dropping or collecting.
- **Monitor the workload:** Open Analytics Dashboard to see the status pie chart, the total number of units and the number ready to collect.

D.3 Instructions for Customers

- Hand over the watch at the service desk and give the staff your name, e-mail address, phone number, the watch brand and model, and a short description of the problem.
- After the booking is saved, check your e-mail for the message titled "Watch Service Appointment Confirmation" with your watch details.
- The service centre will contact you about your appointment and when the watch is ready for pickup.

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APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Register No.	Project Title	Date
THARUN KRISHNA SHANMUGA SUNDAR	241401118	Watch Service Center – Service Booking & Repair Tracking System	02 November 2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner
Name: _____

Signature of External Examiner
Name: _____

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APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Register No.	Department / Section	Date of Viva
THARUN KRISHNA SHANMUGA SUNDAR	241401118	COMPUTER SCIENCE AND BUSINESS SYSTEMS	02 November 2026

Application URL Verified: Yes / No Demo Duration: _____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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